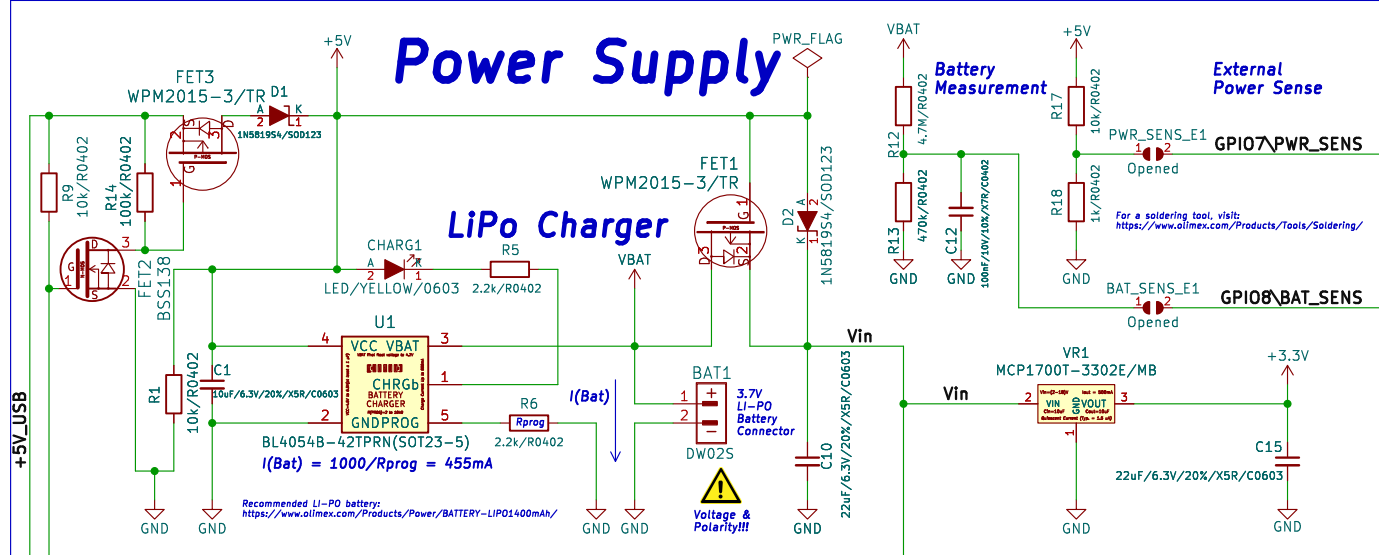


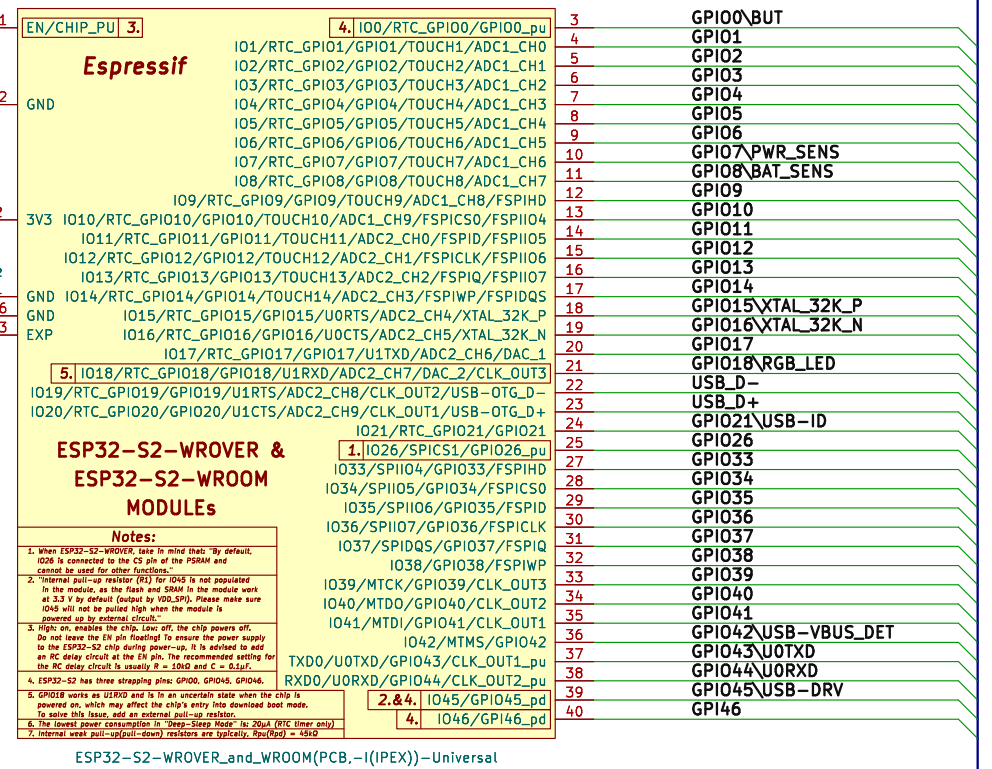
Power Supply



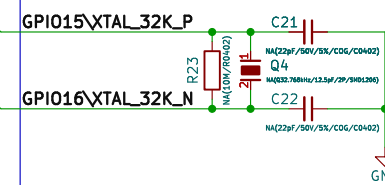
ESP32-S2-xxx_MODULE



U4

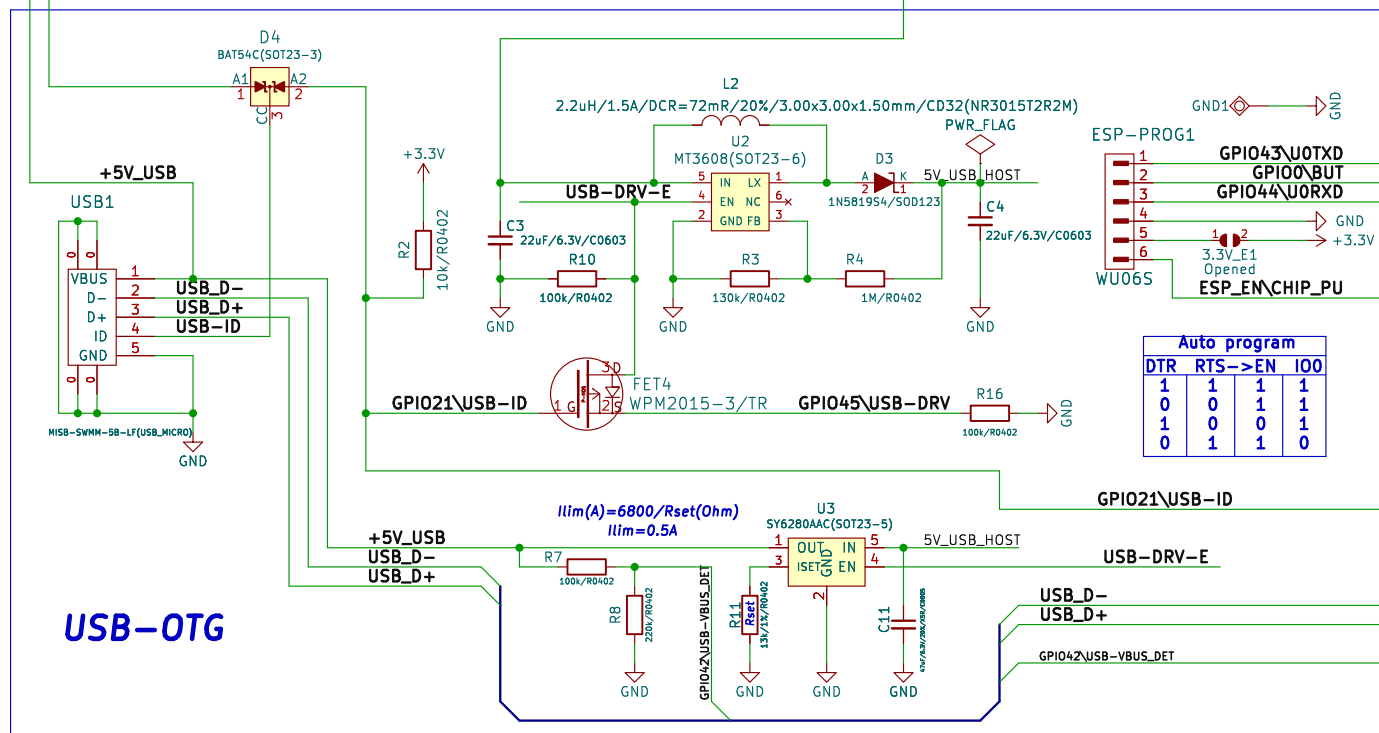


RTC Crystal Option!

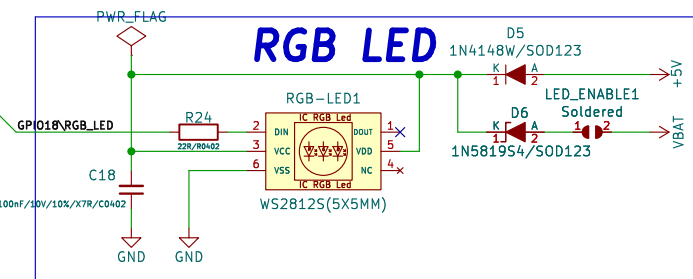


Auto program			
DTR	RTS	EN	I00
1	1	1	1
0	0	1	1
1	0	0	1
0	1	1	0

USB-OTG



RGB LED



Fiducials:



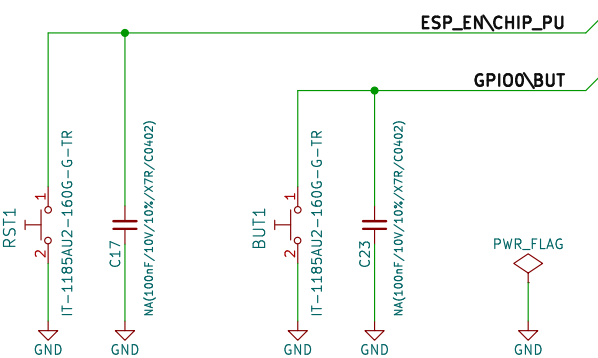
UART_PRINT_CONTROL	GPIO46	ROM Code Printing Control
0	-	ROM code will always print information to UART during boot. GPIO46 is not used.
1	0	Print is enabled during boot
1	1	Print is disabled
2	0	Print is disabled
2	1	Print is enabled during boot
3	-	Print is always disabled during boot. GPIO46 is not used.

ESP32-S2 has three strapping pins: GPIO0, GPIO45, GPIO46

Voltage of Internal LDO (VDD_SPI Voltage)		(Note Note 2)	
Pin	Default	3.3V	1.8V
I045\GPIO45	Pull-Down	0	1
Bootstrapping Mode			
Pin	Default	SPI Flash Boot	Download Boot
I00\GPIO0	Pull-Up	1	0
I046\GPIO46	Pull-Down	Don't-care	0
Enabling/Disabling ROM Code Print During Booting (Note Note 3,4)			
Pin	Default	Enabled	Disabled
I046\GPIO46	Pull-Down	See the fourth note	See the fourth note

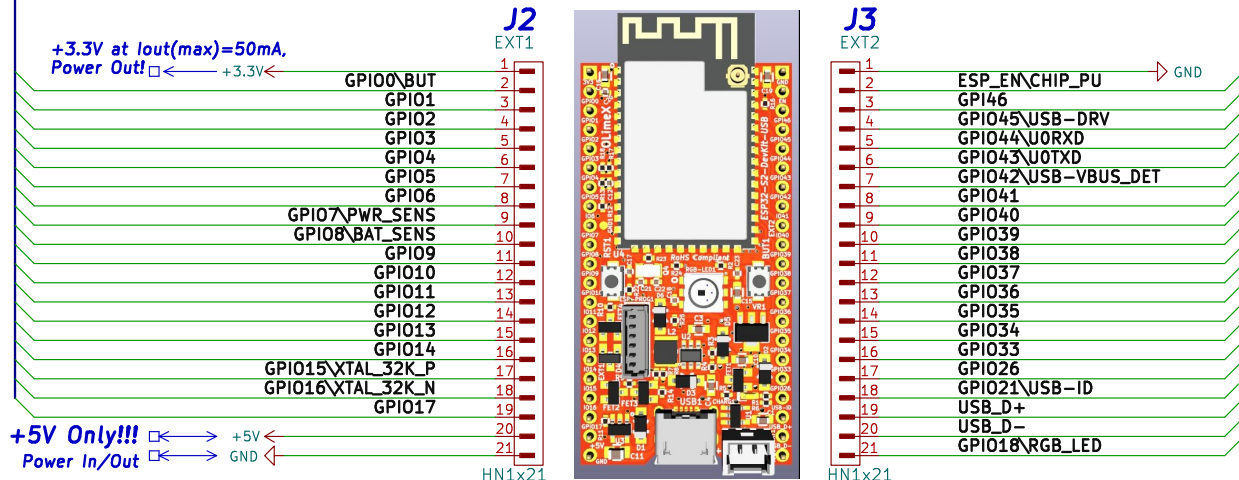
- Notes:**
- Firmware can configure register bits to change the settings of "VDD_SPI Voltage".
 - Internal pull-up resistor (R1) for I045 is not populated in the module, as the flash and SRAM in the module work at 3.3V by default (output by VDD_SPI). Please make sure I045 will not be pulled high when the module is powered up by external circuit.
 - ROM code can be printed over TXD0 (by default) or DAC_1 (I017), depending on the eFuse bit.
 - When eFuse UART_PRINT_CONTROL value is:
0, print is normal during boot and not controlled by I046.
1 and I046 is 0, print is normal during boot; but if I046 is 1, print is disabled.
2 and I046 is 0, print is disabled; but if I046 is 1, print is normal.
3, print is disabled and not controlled by I046.

Reset&User Buttons



Extensions:

Fully compatible to Espressif's
ESP32-S2-SA0LA-1 connectors J2&J3!



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Title: ESP32-S2-DevKit-USB

Size: A3 Date: 2020-11-11
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Rev: B
Id: 1/1 PB-Free